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Treatment pad for a dielectric barrier discharge plasma treatment

Abstract

A treatment pad for a dielectric barrier discharge plasma treatment of a tissue to be treated of an electrically conducting body, which tissue is used as a counter electrode, said treatment pad comprising: a treatment zone, arranged for at least covering the tissue to be treated; a pattern of one or more active areas, integrated in the treatment zone and arranged for generating a dielectric barrier discharge plasma, each said one or more active areas comprising: a first electrode to be coupled to a high voltage power source; a dielectric formed by a coating or foil of a flexible material so that the dielectric shields the first electrode from the tissue to be treated; and a spacer comprising a structured surface of protrusions adjacent a side of the dielectric facing the tissue to be treated; wherein the treatment zone comprises a complementary pattern of non-active areas, such that a complementary part of the tissue to be treated is covered by the said one or more active areas when the treatment pad is reapplied on the object with an offset.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This patent application is a U.S. National Phase of PCT International Application No. PCT/NL2021/050775, filed Dec. 17, 2021, which claims priority to The Netherlands Application No. 2027148, filed Dec. 17, 2020, which are both expressly incorporated by reference in their entireties, including any references contained therein.

FIELD OF THE INVENTION

(2) The invention relates to devices for generating non-thermal plasma, in particular, that can be applied for treatment of living tissue.

BACKGROUND OF THE INVENTION

(3) Cold plasmas have considerable potential for skin conditioning, disinfection of skin and wound healing. From U.S. Pat. No. 9,005,188, EP2670477 and EP3297724 dielectric barrier discharge plasma treatment pads are known, having flexible electrodes to be able to treat an irregularly shaped tissue with plasma. However, these treatment pads are likely only suitable for treating relatively small wounds.

(4) To treat a larger wound, use of several distinct pads can be impractical because of difficulties with proper attachment of the pad to the wound and the risk that parts of the wound will be overtreated or not treated at all.

(5) Simply increasing the pad size may also lead to difficulties because a larger electrode causes more capacitive loading. This results in a lower voltage, which can lead to failure to ignite the plasma and/or inhomogeneous plasma.

(6) To obviate this, one may use a different power source. However, this would introduce new challenges to maintain compliance with electrical safety requirements. For example, the dielectric barrier in the pad and the high voltage cables may need a higher breakdown voltage, or PCB traces in the power source may need to be spaced further apart.

(7) Such changes could increase cost price of the pad quite significantly or make the material much less flexible, hence unusable for its purpose. Thus, some of these changes are challenging and it may not be possible to establish compliance with the electrical safety requirements for the entire system.

(8) The present invention focuses on a cold plasma device which can treat a larger tissue surface, without the drawbacks described.

SUMMARY OF THE INVENTION

(9) In summary, embodiments of the invention pertain to a treatment pad for a dielectric barrier discharge plasma treatment of a tissue surface to be treated of an electrically conducting body, which tissue is used as a counter electrode. The treatment pad comprises a treatment zone, arranged for at least covering the tissue to be treated, with a pattern of one or more active areas integrated in the treatment zone and arranged for generating a dielectric barrier discharge plasma.

(10) Each of the one or more active areas comprises a first electrode to be coupled to a high voltage power source, a dielectric formed by a coating or foil of a flexible material so that the dielectric shields the first electrode from the tissue to be treated, and a spacer comprising a structured surface of protrusions adjacent a side of the dielectric facing the surface to be treated.

(11) The treatment zone comprises a complementary pattern of non-active areas, such that a complementary part of the tissue to be treated is covered by the said one or more active areas when

the treatment pad is reapplied on the object with an offset.

(12) For example, the treatment pad may be applied on the tissue surface to be treated, with the treatment zone covering the entire tissue surface to be treated. The treatment zone comprises a pattern of one or more active areas, e.g. arranged for providing plasma treatment to a first part of the surface to be treated. The treatment zone further comprises a complementary pattern of one or more non-active areas, e.g. arranged for covering, yet not providing any treatment to, a second part of the tissue surface to be treated. The complementary pattern of non-active areas is complementary to the pattern of active areas in that, combined, the pattern of non-active areas and the pattern of active areas may fill out or complete the treatment zone. Moreover, the pattern of active areas and the complementary pattern of non-active areas are complementary in that they are transposable relative to the surface to be treated by reapplying the treatment pad with an offset, without the risk of undertreatment or overtreatment of the surface to be treated, by minimizing the overlap between patterns of active areas after reapplying the treatment pad. The treatment zone may be bounded by an inactive border portion forming an edge of the treatment pad.

(13) For example, by reapplying the treatment pad with an offset, e.g. rotating the treatment pad by 180 degrees in plane of the treatment zone, the treatment zone can again be arranged for covering the entire surface to be treated, yet with a different relative orientation. Now, the pattern of active areas may cover the second part of the surface to be treated, and the complementary pattern of non-active areas may cover the first part of the surface to be treated. As a result, the entire surface to be treated can be covered and treated by the pattern of active areas in multiple stages, e.g. two or more stages, which allows the use of relatively small active areas within a relatively large treatment zone, to treat a large surface to be treated.

(14) By having a treatment pad with a treatment zone that comprises a complementary pattern of non-active areas, such that a complementary part of the surface to be treated is covered by the one or more active areas when the treatment pad is reapplied on the object with an offset, the treatment pad can provide dielectric barrier discharge plasma treatment to relatively large surfaces, without the risk that parts of the surface are undertreated or overtreated and without the effects of capacitive loading.

(15) Surprisingly, the present invention has found a solution that does not require changes to the flexible electrode design and does not require a different electrode design. The innovative concept makes use of active areas in a non-obvious way, by using a pattern of one or more active areas and a complementary pattern of non-active areas to treat a complementary part of the surface to be treated by reapplying the pad with an offset.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The invention will be further elucidated in the figures:
- (2) FIG. 1 illustrates an isometric view of a treatment pad;
- (3) FIG. 2 illustrates a detailed view of a pattern of active areas;
- (4) FIG. 3 illustrates a detailed view of another pattern of active areas;
- (5) FIG. 4 illustrates a detailed view of another pattern of active areas;
- (6) FIG. 5 illustrates a detailed view of another pattern of active areas;
- (7) FIG. 6 illustrates further features of the treatment pad;
- (8) FIG. 7 illustrates a bottom view of the treatment pad;
- (9) FIG. 8 illustrates a control unit (500) for controlling the treatment pad;
- (10) FIG. 9 illustrates a control scheme for the control unit of FIG. 8.

DETAILED DESCRIPTION

- (11) Aspects of the invention relate to a dielectric barrier discharge plasma treatment pad with a

treatment zone having a pattern of one or more active areas and a complementary pattern of non-active areas, such that a complementary part of a surface to be treated is covered by the one or more active areas when the treatment pad is reapplied on the object with an offset, e.g. a translation or rotation in plane of the treatment zone. The benefit is that in this way relatively large surfaces can be treated, with the advantages of having relatively small and flexible electrodes.

(12) With the pattern of one or more active areas and the complementary pattern of non-active areas forming an alternating pattern, a relatively large degree of variations in shapes of the treatment zone, pattern shapes and types of offsets can be used to provide plasma treatment to a surface to be treated. This may provide the user or clinician with treatment options tailored to the need of the patient.

(13) In some preferred embodiments, the pattern comprises one or more elongate active areas and an equal number of non-active areas, e.g. forming a parallel pattern. The elongate active areas have a long side oriented along a lateral direction and a short side oriented along a horizontal direction. The active areas and non-active areas are preferably arranged serially along the horizontal direction. By this, a minimal creepage distance between active areas can be ensured to comply with electrical safety requirements. Additionally, two or more elongate active areas can be equal in size, to have uniform treatment characteristics of active areas along the treatment zone.

(14) Preferably, the treatment pad is reapplied on the object with an offset being a rotation in plane of the treatment zone of one hundred and eighty degrees, e.g. for ease-of-use of the clinician.

(15) In preferred embodiments, the treatment pad further comprises a set of reference markers, each arranged for marking a position on a side of the surface to be treated. The treatment pad can comprise flaps symmetrically extending outward from the treatment zone in a border portion the treatment pad, to engage with the set of reference markers. The reference markers can for example be circular reference stickers with a radius of curvature matching a radius of curvature of cylindrically rounded notches on the flaps of the pad, to provide a means for realigning the treatment pad, such that the complementary part of the surface to be treated is covered by the one or more active areas when the treatment pad is reapplied on the object with an offset.

(16) In some embodiments, the treatment pad further comprises a pad sticker having an adhesive material to attach the treatment pad to the object, e.g. a patient or surface to be disinfected or sterilized. Preferably, an outer contour of the pad sticker at least exposes the set of reference markers, to enable to visually check the alignment of the treatment pad on the object after it has been attached.

(17) In other or further embodiments, the treatment pad comprises an edge around a circumference of the treatment zone, to create a sealed contact between the treatment pad and the object to prevent airflow between the surroundings and the active areas and to improve electrical safety of the pad.

(18) Other aspects of the invention pertain to a control unit for controlling the treatment pad as disclosed herein, comprising a controller and a high voltage power source for controlling the voltage to the one or more active areas. The controller is arranged for activating a pattern of one or more active areas when the treatment pad is first applied on the object, and for activating a superset, set or subset of active areas in the pattern of one or more active areas such that a complementary part of the surface to be treated is covered by the subset of active areas when the treatment pad is reapplied on the object with an offset. In case of partial overlap or misalignment of the pattern, this allows activating only those active areas that form the complementary part of the surface to be treated, to avoid the risk that parts of the surface are undertreated or overtreated. A superset may encompass the set or subset.

(19) By having the controller additionally arranged for sequentially activating active areas, exceeding a total output voltage limit of the treatment pad can be avoided. Additionally the controller can be arranged for having the pattern of one or more active areas provide a dielectric barrier plasma to the surface to be treated with a predefined first duration and intensity when the treatment pad is first applied on the object, and for having the pattern of one or more active areas

provide a dielectric barrier plasma to the surface to be treated with a matching second duration and intensity when the treatment pad is reapplied on the object with an offset. This promotes that all parts of the surface to be treated are provided with an equal dose of plasma.

(20) The invention is described more fully hereinafter with reference to the accompanying drawings, in which embodiments of the invention are shown. In the drawings, the absolute and relative sizes of systems, components, layers, and regions may be exaggerated for clarity. Embodiments may be described with reference to schematic and/or cross-section illustrations of possibly idealized embodiments and intermediate structures of the invention. In the description and drawings, like numbers refer to like elements throughout. Relative terms as well as derivatives thereof should be construed to refer to the orientation as then described or as shown in the drawing under discussion. These relative terms are for convenience of description and do not require that the system be constructed or operated in a particular orientation unless stated otherwise.

(21) FIG. 1 illustrates an isometric view of a treatment pad **10** for a dielectric barrier discharge plasma treatment of a surface **5** to be treated of an electrically conducting body, which surface **5** is used as a counter electrode. In a preferred embodiment, the treatment pad **10** comprises a treatment zone **20**, arranged for at least covering the surface **5** to be treated. The treatment zone **20** can be rectangular, as shown in FIG. 1, or a different geometrical or irregular shape that covers the surface **5** to be treated. The treatment pad **10** further comprises a pattern of one or more active areas **30**, integrated in the treatment zone **20** and arranged for generating a dielectric barrier discharge plasma. Preferably, each of the one or more active areas **30** comprises a first electrode **31**, a dielectric **32**, and a spacer **33**. The first electrode **31** is to be coupled to a high voltage power source. This may be carried out simultaneously for all active areas in the treatment pad, or may be carried out sequentially; e.g. by manually connecting a high voltage source to an electrode **31** via a terminal clamp or the like. In other embodiments, active areas may be interconnected via connecting conductors integrated in the treatment pad or each active area may have an individual connecting terminal pin. The dielectric **32** is formed by a coating or foil of a flexible material so that the dielectric **32** shields the first electrode **31** from the surface **5** to be treated. The spacer **33** comprises a structured surface of protrusions adjacent a side of the dielectric **32** facing the surface **5** to be treated. In some embodiments, the dielectric **32** and spacer **33** span at least part of the treatment zone **20** and are shared by multiple active areas **30**, while each active area **30** comprises an individual first electrode **31**. By each having a first electrode **31**, the active areas **30** can be activated independently from each other, for example by connecting, in a specific order, first electrodes **31** of active areas **30** to a high voltage power source. Alternatively, multiple active areas **30** can be activated simultaneously, by having their first electrodes **31** simultaneously powered by a high voltage power source. Activating the active areas **30** and/or setting the order in which the active areas **30** are activated can be a manual process or a (semi)automated process. In a preferred embodiment, the treatment zone **20** comprises a complementary pattern of non-active areas, such that a complementary part of the surface **5** to be treated is covered by the said one or more active areas **30** when the treatment pad **10** is reapplied on the object with an offset.

(22) The benefit of a treatment pad **10** such as the embodiment shown in FIG. 1 is that it can provide relatively large surfaces with dielectric barrier discharge plasma treatment, without the risk that parts of the surface are undertreated or overtreated and without the effects of capacitive loading.

(23) In some embodiments, as shown in FIGS. 2-5, the treatment pad **10** comprises a treatment zone **20** with a pattern of one or more active areas **30** and a complementary pattern of non-active areas forming an alternating pattern of active areas **30** and non-active areas. The purpose of the embodiments shown in FIGS. 2-5 is to schematically illustrate potential variations of these patterns, without being limited to the specific pattern as depicted. Also, obviously, realistic patterns are constrained by plasma forming properties e.g. regarding sharp edges. Also, no specific format is provided for the active area, as long as it is able to provide the plasma forming effect.

(24) FIG. 2 illustrates a detailed view of a pattern of active areas **30**. As shown in FIG. 2A, the pattern comprises one or more elongate active areas **30** and an equal number of non-active areas. By having an equal number of active areas **30** and non-active areas, a complementary part of the surface **5** to be treated can be covered by the active areas **30** when the treatment pad **10** is reapplied on the object with an offset, without active areas **30** covering a part of the object beyond the surface **5** to be treated. Accordingly, the size of the treatment pad **10** can be closely matched with the size of the surface **5** to be treated.

(25) Alternatively, the pattern can comprise an unequal number of active areas **30** and non-active areas. For patterns comprising more active areas **30** than non-active areas, a number of active areas **30** would cover a part of the object beyond the surface **5** to be treated when the treatment pad **10** is reapplied on the object with an offset. To avoid providing plasma to parts of the object beyond the surface **5** to be treated, a superset or subset of the pattern of active areas **30** covering the surface **5** to be treated can be activated, while the remaining active areas **30** (not covering the surface **5** to be treated) are deactivated.

(26) For patterns comprising fewer active areas **30** than non-active areas, the treatment pad **10** can be reapplied on the object with an offset multiple time, e.g. an incremental offset, such as a stepwise rotation or translation or combination thereof, until the entire surface **5** to be treated has been covered by the pattern of active areas **30**.

(27) As shown in FIG. 2, the active areas **30** have a long side oriented along a lateral direction Y and a short side oriented along a horizontal direction X. The active areas **30** and non-active areas are arranged serially along the horizontal direction X, e.g. forming a parallel pattern. The orientation of treatment zone **20** does not necessarily need to be aligned with the orientation of active areas **30**.

(28) Alternatively, the active areas **30** can have a long side oriented along the horizontal direction X and a short side oriented along the lateral direction Y.

(29) Additionally, or alternatively, the elongate active areas **30** can be curved, e.g. have a sinusoidal shape with peaks forming the short side, e.g. waves moving in the horizontal direction X.

(30) A potential benefit of a parallel pattern is that elongate active areas **30** are spaced apart with a uniform interdistance along their long side, which can ensure a minimal creepage distance between active areas **30** to comply with electrical safety requirements, such as IEC 60601-1 for medical devices.

(31) In a preferred embodiment, two or more elongate active areas **30** are equal in size. For example, in a pattern comprising more than one active area **30**, all active areas **30** can be equal in size, or a subset of all active areas **30** can be equal in size, or at least two active areas **30** can be equal in size. Having active areas **30** equal in size promotes equal operating characteristics, such as the generation of plasma and the effects of capacitive loading, which can be important usability, safety or control parameters.

(32) Preferably, the treatment pad **10** is reapplied on the object with an angular offset, being a rotation in plane of the treatment zone **20** of one hundred and eighty degrees. This has been found to be most user-friendly (to the clinician), because of a relatively easy re-alignment procedure.

(33) FIG. 2A shows the pattern of elongate active areas **30** and the complementary pattern of active areas **30'** when the treatment zone **20'** is reapplied on the object with an angular offset DR of one hundred and eighty degrees around a central origin **38**.

(34) Accordingly, the rotated treatment zone **20'** is realigned with the original treatment zone **20**. As a result, the surface **5** to be treated can effectively be equal to or smaller than the treatment zone **20**.

(35) The treatment zone **20** can alternatively be a different geometrical or irregular shape and the offset can alternatively be a translation of the pattern along the short side of the active areas **30**, though this may reduce the effective size of the surface **5** to be treated.

(36) In FIG. 3A, the treatment zone **20** is rectangular and comprises a two-dimensional alternating pattern of a plurality of rectangularly shaped active areas **30** and complementary non-active areas,

e.g. forming a checkerboard pattern.

(37) In some embodiments, the first electrodes **31** of adjacent active areas **30** are electrically isolated from each other and each of the active areas **30** comprises an electrode connector to connect its first electrode **31** to a high voltage power source for independent activation of active areas **30**.

(38) Alternatively, the first electrodes **31** of selected active areas **30** can be electrically connected to each other, e.g. to form a predefined subset of active areas **30** comprising a shared electrode connector for combined activation of active areas **30**. The predefined subset of electrically connected active areas **30** may comprise active areas **30** that are topographically separated, e.g. having at least a horizontal distance of $2W$ in the horizontal direction X and a lateral distance $2H$ in the lateral direction Y between active areas **30** in a subset.

(39) The complementary part of the surface **5** to be treated, shown in FIG. **3B**, can be covered by translating the treatment zone **20** along the lateral direction DY . When the treatment zone **20'** is reapplied on the object, the active areas **30'** are shifted along the lateral direction DY by a height H of an active area **30**. This creates a row of alternating active areas **30**, **30'** and non-active areas on two opposing edges of the surface **5** to be treated. As a result, the surface **5** to be treated is effectively smaller than the treatment zone **20**.

(40) With respect to the embodiment of FIG. **3**, steps for providing plasma treatment to the surface **5** to be treated can e.g. be as follows. At the start of the procedure the treatment zone **20** is first applied on an object, with a first set or subset of the pattern of active areas **30** covering a first part of the surface **5** to be treated. Specific sets or subsets of active areas **30** can e.g. be activated by having active areas **30** of which the first electrodes **31** are electrically isolated from each other and that comprise individual electrode connectors to connect the active areas **30** to a high voltage power source. As shown in FIG. **3A**, the first subset comprises the top three rows of active areas **30** in treatment zone **20**. Accordingly, the first subset of active areas **30** is activated to provide a dielectric barrier discharge plasma to the first part of the surface **5** to be treated. In the checkerboard pattern, active areas that are adjacent may be activated sequentially e.g. to limit capacitive loading in the active areas.

(41) Next, the treatment zone **20** is reapplied on the object with a translation along DY by a height H of an active area. This causes a second subset of the pattern of active areas **30'** to cover a second part of the surface **5** to be treated, complementary to the first part of the surface **5** to be treated. As shown in FIG. **3B**, the second subset comprises the bottom three rows of active areas **30'** in treatment zone **20'**. Finally, the second subset of active areas **30'** is activated to provide a dielectric barrier discharge plasma to the complementary part of the surface **5** to be treated. Also, here, active areas may be activated sequentially or simultaneously if possible.

(42) Alternatively, the complementary part of the surface **5** to be treated can be covered by translating the treatment zone along the horizontal direction, perpendicular to the lateral direction DY in plane of the treatment zone **20**. Accordingly, the treatment zone **20'** is reapplied on the object, with the active areas **30'** shifted by a width W of an active area **30**.

(43) The treatment zone **20** can alternatively be a different geometrical or irregular shape, and the checkerboard pattern of active areas **30** can comprise an odd or even number of rows or columns.

(44) Additionally, or alternatively, the orientation of the active areas **30** can be tilted relative to the orientation of the treatment zone **20**, and within the treatment zone **20** the active areas **30** may vary in size.

(45) FIG. **4** illustrates a detailed view of another pattern of active areas **30**. FIG. **4A** shows a circular treatment zone **20** with an alternating pattern of active areas **30** and complementary non-active areas aligned with respect to a central origin **38**, e.g. forming a cloverleaf pattern. With this pattern, the complementary part of the surface **5** to be treated can be covered by the active areas **30'** by rotating the treatment zone **20** along an angular offset DR in plane of the surface **5** to be treated of ninety degrees, either clockwise or counter-clockwise. When the treatment zone **20'** is reapplied

on the object, as shown in FIG. 4B, the rotated treatment zone **20'** is aligned with the original treatment zone **20**. As a result, the surface **5** to be treated can effectively be equal to or smaller than the treatment zone **20**.

(46) Alternatively, the pattern of active areas **30** shown in FIG. 4A can comprise less or more active areas **30** and complementary non-active areas. Alternatively, the pattern can comprise regularly or irregularly divided circle segments, with or without notches, e.g. forming a dartboard pattern. In some embodiments, the active areas **30** comprise first electrodes **31** that are electrically isolated from each other, e.g. to allow activation of subsets.

(47) In the embodiment shown in FIG. 5, the treatment zone **20** is elongated with rounded ends, and comprises an alternating pattern of active areas **30** and complementary non-active areas, in which the active areas **30** have different shapes and sizes, e.g. forming a band-aid pattern. The active areas **30** comprise rounded edges to follow the contour of the treatment zone **20** and to create a sinusoidal boundary, crossing a central origin **38**, between an active area **30** and a non-active area, as can be seen in FIG. 5A. Active areas **30** may comprise first electrodes **31** that are electrically isolated from each other, e.g. to allow activation of subsets.

(48) FIG. 5B shows the treatment zone **20'** when the treatment pad **10** is reapplied on the object with an angular offset DR. The original treatment zone **20** can be rotated around the central origin **38** by one hundred and eighty degrees and reapplied on the surface **5** to be treated. The corresponding complementary pattern of active areas **30'** on the rotated treatment zone **20'** is able to entirely cover a surface **5** to be treated that is equal to or smaller than the treatment zone **20**.

(49) FIG. 6 illustrates further features of the treatment pad **10**. As shown, the treatment pad **10** further comprises a set of reference markers **70**, each arranged for marking a position on a side of the surface **5** to be treated. The reference markers **70** can for example be placed on opposing sides of the surface **5** to be treated, as shown in FIG. 6. By marking a position on a side of the surface **5** to be treated, the treatment pad **10** can be realigned with respect to the reference markers **70** when the treatment pad **10** is reapplied on the object with an offset.

(50) Alternatively, the set of reference markers **70** can be arranged on adjacent sides of the surface **5** to be treated, or the set of reference markers **70** can be arranged on one side of the surface **5** to be treated. However, it is recommended that the marked distance between reference markers **70** is as large as practically possible, because this decreases the risk of alignment error when the treatment pad **10** is reapplied on the object.

(51) In another or further embodiment, a first reference marker **70** can be arranged for marking a position corresponding to a center of rotation for an angular offset of the treatment pad **10**. A second and third reference marker **70** can be used for defining an initial orientation of the treatment pad **10**, and an orientation of the treatment pad **10** when it is reapplied on the object with an angular offset, respectively.

(52) Preferably, the treatment pad **10** comprises flaps **80** symmetrically extending outward from the treatment zone **20** to engage with the set of reference markers **70**. For example, as shown in FIG. 6 and FIG. 7, the flaps **80** extend from opposing sides of the treatment zone **20** toward the reference markers **70**.

(53) Alternatively, in combination with sets of reference markers **70** prescribed to be arranged e.g. on adjacent sides or on one side, the flaps **80** can extend from sides of the treatment zone **20** that correspond with the prescribed position of the reference markers **70**.

(54) In some embodiments, for example as shown in FIG. 6, the reference markers **70** are circular reference stickers with a radius of curvature matching a radius of curvature of cylindrically rounded notches **75** on the flaps **80** of the pad **10**. The matching radii of curvature can thus be used to realign the treatment pad **10** with the reference markers **10** when the treatment pad **10** is reapplied on the object with an offset.

(55) Alignment can for example be done visually. In other embodiments, alignment is done mechanically by having reference markers **70** and flaps **80** with a substantial thickness, e.g. a

thickness at least two millimeters, preferably between two and five millimeters. When the reference marker **70** and the flap **80** are mechanically aligned, this can additionally provide in-plane stability of the treatment pad **10** relative to the object during a dielectric barrier discharge plasma treatment session, which may reduce the risk that parts of the surface are undertreated or overtreated.

(56) Alternatively, mechanical alignment between the reference markers **70** and the treatment pad **10** can comprise engaging elements, e.g. forming a click system. For example, the reference markers **70** can have a mushroom shape, i.e. with a base portion having a relatively small diameter which is attached to the skin of the patient, and an upper portion having a relatively large diameter. The diameter of the base portion can be used to align the reference marker **70** with a cylindrically rounded notch **75** in plane of the treatment zone relative to the object, and the diameter of the upper portion protruding over the flap **80** can constrain vertical movement of the treatment zone relative to the object.

(57) Instead of being circular, the reference markers **70** can comprise a different predefined geometrical shape, e.g. arrow, diamond, hexagon, triangle or rectangle, matching a notch **75** on a flap **80** on the treatment pad **10**.

(58) Alternatively, the flap **80** may extend beyond the reference marker **70**. Instead of a notch **75**, the flap **80** may comprise an indent from a bottom surface of the flap **80**, with the indent having a shape matching the shape of the reference marker **70** to provide an alignment means. The indent preferably has a height larger than the thickness of the reference marker **70**. Alternatively, the indent can be a through-hole, running from a bottom surface of the flap **80** to a top surface of the flap **80**.

(59) In another or further preferred embodiment, the treatment pad **10** comprises a pad sticker **90** having an adhesive material to attach the treatment pad **10** to the object, as can be seen in FIG. 6. For example, the pad sticker **90** has a central cutout to expose the projected top of the treatment zone **20** and to allow access to the electrode connectors **40**. In some embodiments, the outer contour of the pad sticker **90** at least exposes the notches **75** on the flaps **80**, so that alignment of the treatment pad **10** with the reference markers **70** can be checked visually. Preferably, the pad sticker **90** does not cover or adhere to the reference markers **70**, to improve usability and to avoid removing the reference markers **70** from the skin of the patient when the treatment pad is reapplied with an offset.

(60) Alternatively, the pad sticker **90** covers the reference markers **70** and is at least partially transparent to allow visual alignment of the treatment pad **10** with the reference markers **70**.

(61) As shown in FIG. 6, an inner portion of the pad sticker **90** is attached to a top surface of the treatment pad **10**, and an outer portion of the pad sticker **90** is attached to the object, e.g. the skin of a patient. Alternatively, the pad sticker **90** can e.g. be a double-sided adhesive sticker attaching a bottom surface of the treatment pad **10** to the object.

(62) Additionally, or alternatively, the pad sticker **90** can be used to attach flaps **80** to the object, e.g. by a double-sided adhesive on a bottom surface of flaps **80**. Alternatively, the functionalities of flaps **80** and pad sticker **90** can be integrated. For example, the flaps **80** can have an adhesive material to attach the treatment pad **10** to the object, or the pad sticker **90** can be an integral part of the treatment pad **10** and have notches **75** with a radius of curvature matching a radius of curvature of circular reference markers **70**.

(63) Preferably, the adhesive material is friendly to sensitive and elderly skin, to avoid irritation or allergic reactions.

(64) Preferably the adhesive material allows multiple reapplications of the pad sticker **90**, to reattach the treatment pad **10** to the object when the treatment pad **10** is reapplied on the object with an offset.

(65) Instead of being a single ring-shaped sticker, as shown in FIG. 6, the pad sticker **90** can comprise a set of multiple stickers to attach the treatment pad **10** to the object, for example having stickers attached to corners or edges of the pad **10**.

(66) FIG. 7 illustrates a bottom view of the treatment pad **10**. In another or further embodiment as shown, the treatment pad **10** further comprises an edge **85** around a circumference of the treatment zone **20**, e.g. a rectangular or circular edge depending on the shape of the treatment zone **20**, to create a sealed contact between the treatment pad **10** and the object to prevent airflow between the surroundings and the active areas **30** and to improve electrical safety of the treatment pad **10**, e.g. to comply to medical device regulations. Each active area **30** is arranged for generating a dielectric barrier discharge plasma and comprises a first electrode **31** (see FIG. 1), a dielectric **32**, and a spacer **33** comprising a structured surface of protrusions. The thickness of the spacer **33** defines the distance between the dielectric **32** and the surface **5** to be treated, to ensure sufficient air is available in the cavity to convert to plasma, to reduce the risk of contact between the dielectric **32** and the surface **5** to be treated, and to reduce the risk of wound fluid filling the cavity. A larger thickness of the spacer **33** creates a larger offset between the first electrode **31** and the surface **5** to be treated, which in turn may increase the voltage required to generate plasma in the cavity. For example, the spacer **33** has a thickness of more than half a millimeter, preferably between half a millimeter and three millimeters, more preferably between half a millimeter and one and a half millimeter.

(67) As shown in FIG. 7, the edge **85** can for example be an integrally formed part with flaps **80**. Preferably, the edge **85** forms a seal around the treatment zone **20** with a sufficient margin of distance (i.e. a minimum creepage distance) between the first electrode(s) and the edge of the treatment pad for electrical safety of the treatment pad **10**.

(68) In some embodiments, the edge **85** comprises an adhesive material to attach the treatment pad **10** to the object.

(69) Preferably, the distance W between adjacent active areas **30** is at least ten millimeters, preferably between ten and eighty millimeters, most preferably between fifteen and fifty millimeters. This ensures a minimum distance between adjacent active areas **30**, e.g. to make sure an electrode connector of a non-activated active area **30** is safe to touch, which may be required to comply to technical standards such as IEC 60601-1 for medical devices. Alternatively, when the active areas **30** are provided with additional electrical isolation means, the minimum distance between active areas **30** can be smaller. In some embodiments, an additional distance of e.g. between one and three millimeters, preferably one millimeter, is added between active areas **30** and non-active areas, to allow for manual placement tolerances when repositioning the pad.

(70) Additionally, this provides a scalable pattern of active areas **30** in which the size of active areas **30** limits capacitive loading. Thus, when treating relatively large surfaces **5**, the present invention may not require changes to the output voltage of the treatment pad **10** to successfully ignite the plasma.

(71) This may offer perspectives for medical treatments and prevention measures for larger wounds. In dermatology, new opportunities can be opened for wound healing, tissue regeneration, therapy of skin infections, and probably many more applications.

(72) In preferred embodiments, the treatment pad **10** comprises a treatment zone **20** arranged for covering a surface **5** to be treated that is larger than one thousand two hundred millimeter squared. For example, a surface **5** to be treated between one thousand two hundred millimeter squared and sixty thousand millimeter squared, preferably between two thousand four hundred millimeter squared and ten thousand millimeter squared.

(73) These large surface areas can be found in the treatment of skin diseases, such as bacterial or fungal infections, or chronic wounds and inflammations. Plasmas may very well assist in controlling the consequences of chronic inflammation associated with these diseases by eliminating bacterial and fungal infections, which results in a drastic improvement of the quality of life.

(74) Secondly, having these large surface areas available for dielectric barrier discharge plasma treatment may benefit hospital hygiene by sterilizing or disinfecting medical equipment and body parts (e.g. of surgeons).

(75) FIG. 8 illustrates a control unit **500** for controlling the treatment pad **10**, with a control scheme for the control unit **500** shown in FIG. 9. In a preferred embodiment, the control unit comprises a controller **520** and a high voltage power source **510** for controlling the voltage to the one or more active areas **30**. Preferably, the controller **520** is arranged for activating a first subset of active areas **30** in the pattern of one or more active areas **30** when the treatment pad **10** is first applied on the object, e.g. a number of active areas **30** that cover a specific area of the surface **5** to be treated are initially activated from the pattern of active areas **30**. Preferably, the controller **520** is further arranged for activating a second subset of active areas **30** in the pattern of one or more active areas **30** such that a complementary part of the surface **5** to be treated is covered by the subset of active areas **30** when the treatment pad **10** is reapplied on the object with an offset. The second subset e.g. is a number of active areas **30** that cover another area of the surface **5** to be treated, complementary to the initially covered area of the surface **5** to be treated.

(76) This can for example be used when the offset is a translation and a number of active areas **30** do not cover the surface **5** to be treated, as illustrated in FIG. 3.

(77) More than two subsets of active areas **30** may be required to cover the entire surface **5** to be treated. Accordingly, the treatment pad **10** may be reapplied on the object an equal number of times.

(78) Alternatively, different subsets of active areas **30** can be activated in a number of treatment steps while the treatment pad **10** remains in position on the object to be treated, e.g. to limit capacitive loading by having smaller subsets of active areas **30**, or to treat larger wounds without being limited by capacitive loading.

(79) In some embodiments, the control unit **500** comprises a controller **520** that is additionally arranged for sequentially activating active areas **30** to avoid exceeding a total output voltage limit of the treatment pad **10**.

(80) The controller can be arranged to activate active areas **30** in a predefined optimal order, e.g. to limit capacitive loading, to limit local increase in temperature of the treatment pad **10** or of the surface **5** to be treated, or to improve treatment effectiveness. Alternatively, or additionally, the predefined optimal order of activating the active areas **30** can be based on the layout of the wound or the anatomy of the affected body part, e.g. by activating a subset of active areas **30** that cover the surface **5** to be treated.

(81) In other or further embodiments, the control unit **500** comprises a controller **520** that is additionally arranged for having the pattern of one or more active areas **30** provide a dielectric barrier plasma to the surface **5** to be treated with a predefined first duration and intensity when the treatment pad **10** is first applied on the object, and for having the pattern of one or more active areas **30** provide a dielectric barrier plasma to the surface **5** to be treated with a matching second duration and intensity when the treatment pad **10** is reapplied on the object with an offset. This can promote that the plasma dose is equally distributed to the surface **5** to be treated.

(82) Alternatively, the controller **520** can be arranged for having the second duration and/or intensity differ from the first duration and/or intensity, e.g. to provide a varying dose of plasma treatment, based on varying degrees of disease of regions within the wound.

(83) FIG. 9 illustrates a control scheme for the control unit **500** of FIG. 8. The procedure is started by a user (e.g. a clinician) first applying the treatment pad **10** on an object. Next, the control unit **500** is provided with a signal to activate the pattern of one or more active areas **30** (step **501**), to provide a first dielectric barrier discharge plasma treatment to the surface **5** to be treated.

(84) When step **501** is done, the user reapplies the treatment pad **10** on the object with an offset. Next, control unit **500** is provided with a signal to activate a subset in the pattern of one or more active areas **30** to cover a complementary part of the surface **5** to be treated (step **502**).

(85) When step **502** is done, the user verifies if the entire surface **5** to be treated has successfully received plasma treatment. If this is true, the treatment is complete. If this is not true, the steps of reapplying the treatment pad **10** on the object with an offset and step **502** of the control unit **500** are

repeated, until the outcome of the verification step is positive.

(86) The invention also relates to a treatment pad for a dielectric barrier discharge plasma treatment of a surface to be treated of an electrically conducting body, which surface is used as a counter electrode, said treatment pad comprising: a treatment surface, arranged for at least covering the surface to be treated; a pattern of one or more active areas, integrated in the treatment surface and arranged for generating a dielectric barrier discharge plasma, each said one or more active areas comprising: a first electrode to be coupled to a high voltage power source; a dielectric formed by a coating or foil of a flexible material so that the dielectric shields the first electrode from the surface to be treated; and a spacer comprising a structured surface of protrusions adjacent a side of the dielectric facing the surface to be treated;

wherein the treatment surface comprises a complementary pattern of non-active areas, is covered by the said one or more active areas when the treatment pad is reapplied on the object with an offset.

Claims

1. A treatment pad for a dielectric barrier discharge plasma treatment of tissue to be treated of an electrically conducting object, wherein the tissue is used as a counter electrode, and wherein the treatment pad comprises: a treatment zone, arranged for at least covering a surface of the tissue to be treated; a pattern of one or more active areas, integrated in the treatment zone and arranged for generating a dielectric barrier discharge plasma to a first part of the tissue to be treated, each of the one or more active areas comprising: a first electrode to be coupled to a high voltage power source; a dielectric formed by a coating or foil of a flexible material so that the dielectric shields the first electrode from the surface of the tissue to be treated; and a spacer comprising a structured surface of protrusions adjacent a side of the dielectric facing a first part of the surface of the tissue to be treated, wherein a complementary pattern of non-active areas is arranged, in the treatment zone, to cover a second part of the surface of the tissue to be treated that is complementary to the first part, and wherein the one or more active areas are arranged, in a condition when the treatment pad is reapplied on the object with an offset, to cover the second part of the surface of the tissue to be treated.

2. The treatment pad according to claim 1, wherein the pattern of one or more active areas and the complementary pattern of non-active areas form an alternating pattern of active areas and non-active areas.

3. The treatment pad according to claim 2, wherein the pattern of one or more active areas and complementary pattern of non-active areas comprises one or more elongate active areas and an equal number of non-active areas, wherein the active areas have a long side oriented along a lateral direction and a short side oriented along a horizontal direction, and wherein the active areas and non-active areas are arranged serially along the horizontal direction.

4. The treatment pad according to claim 1, wherein the pattern of one or more active areas and complementary pattern of non-active areas comprises one or more elongate active areas and an equal number of non-active areas, wherein the active areas have a long side oriented along a lateral direction and a short side oriented along a horizontal direction, and wherein the active areas and non-active areas are arranged serially along the horizontal direction.

5. The treatment pad according to claim 1, wherein two or more elongate active areas are equal in size.

6. The treatment pad according claim 1, wherein the offset is a rotation in plane of the treatment zone of one hundred and eighty degrees.

7. The treatment pad according to claim 1, further comprising a set of reference markers, wherein each reference marker is arranged for marking a position on a side of the surface of the tissue to be treated.

8. The treatment pad according to claim 7, wherein the treatment pad comprises flaps symmetrically extending outward from the treatment zone to engage with the set of reference markers.
 9. The treatment pad according claim 7, further comprising a pad sticker having an adhesive material to attach the treatment pad to the object.
 10. The treatment pad according to claim 9, wherein an outer contour of the pad sticker at least exposes the set of reference markers.
 11. The treatment pad according claim 1, further comprising an edge around a circumference of the treatment zone.
 12. The treatment pad according claim 1, wherein the distance between adjacent active areas is at least ten millimeters.
 13. The treatment pad according claim 1, wherein the treatment zone is arranged for covering a tissue to be treated that is larger than one thousand two hundred millimeter squared.
 14. A control unit for controlling a treatment pad for a dielectric barrier discharge plasma treatment of tissue to be treated of an electrically conducting object, wherein the tissue is used as a counter electrode, wherein the treatment pad comprises: a treatment zone arranged for at least covering a surface of the tissue to be treated; a pattern of one or more active areas, integrated in the treatment zone and arranged for generating a dielectric barrier discharge plasma to a first part of the tissue to be treated, each of the one or more active areas comprising: a first electrode to be coupled to a high voltage power source; a dielectric formed by a coating or foil of a flexible material so that the dielectric shields the first electrode from the surface of the tissue to be treated; and a spacer comprising a structured surface of protrusions adjacent a side of the dielectric facing a first part of the surface of the tissue to be treated, wherein a complementary pattern of non-active areas is arranged, in the treatment zone, to cover a second part of the surface of the tissue to be treated that is complementary to the first part, and wherein the one or more active areas are arranged, in a condition when the treatment pad is reapplied on the object with an offset, to cover the second part of the surface of the tissue to be treated; and wherein the control unit comprises: a controller; and a high voltage power source for controlling the voltage to the one or more active areas, wherein the controller is arranged for: activating a first subset of active areas in the pattern of one or more active areas when the treatment pad is first applied on an object, and activating a second subset of active areas in the pattern of one or more active areas such that a complementary part of the tissue to be treated is covered by the subset of active areas when the treatment pad is reapplied on the object with an offset.
 15. The control unit according to claim 14, wherein the controller is additionally arranged for sequentially activating active areas.
 16. The control unit according to claim 15, wherein the controller is additionally arranged for having the pattern of one or more active areas provide a dielectric barrier plasma to the tissue to be treated with: a predefined first duration and intensity when the treatment pad is first applied on the object, and a predefined second duration and intensity when the treatment pad is reapplied on the object with an offset.
 17. The control unit according to claim 14, wherein the controller is additionally arranged for having the pattern of one or more active areas provide a dielectric barrier plasma to the tissue to be treated with: a predefined first duration and intensity when the treatment pad is first applied on the object, and a predefined second duration and intensity when the treatment pad is reapplied on the object with an offset.
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